

CUSTOMER SEGMENTATION ANALYSIS

Data Science and Machine Learning Project Documentation

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1. Project Objective

The primary objective of this project is to segment a retail mall's customer base into distinct groups based on their core financial profiles and behavior (Annual Income and Spending Score). Using unsupervised clustering, this helps marketing teams target individual buyer categories with bespoke, hyper-targeted campaigns to improve conversions and retention.

2. Technology Stack

- Programming Language:** Python
- Data Manipulation:** Pandas, NumPy
- Data Visualization:** Matplotlib, Seaborn
- Machine Learning:** Scikit-learn (K-Means Clustering)

3. Dataset Overview

The target dataset utilized is `Mall_Customers.csv`, which tracks a total population of **200 rows**. The data contains zero missing or null entries, providing an ideal set for mathematical clustering.

Data Schema and Statistical Bounds:

Column Name	Data Type	Description & Limits
CustomerID	Int64	Unique baseline token assigned to each individual customer profile.
Gender	Object / Str	Categorical attribute tracking customer gender (Female: 112 counts, Male: 88 counts).
Age	Int64	Numerical age feature (Minimum: 18 years, Maximum: 70 years, Average: 38.85 years).
Annual Income (k\$)	Int64	Customer annual purchasing power in thousands of dollars (Range: \$15k to \$137k).

Column Name	Data Type	Description & Limits
Spending Score (1-100)	Int64	A metric scale scaled from 1 to 100 based on internal mall transactions and frequency.

4. Machine Learning Implementation

This pipeline employs the **K-Means Clustering** algorithm, a foundational tool in unsupervised machine learning. To establish an optimal number of clusters (\$K\$), the **Elbow Method** was simulated through Within-Cluster Sum of Squares (WCSS). The clear inflection point on the curve conclusively settled the hyperparameter bounds at **K = 5**.

5. Visual Clustering Matrix

The generated machine learning cluster output map, visualizing the distribution of the five distinct customer groups against their calculated cluster centroids, is presented below:

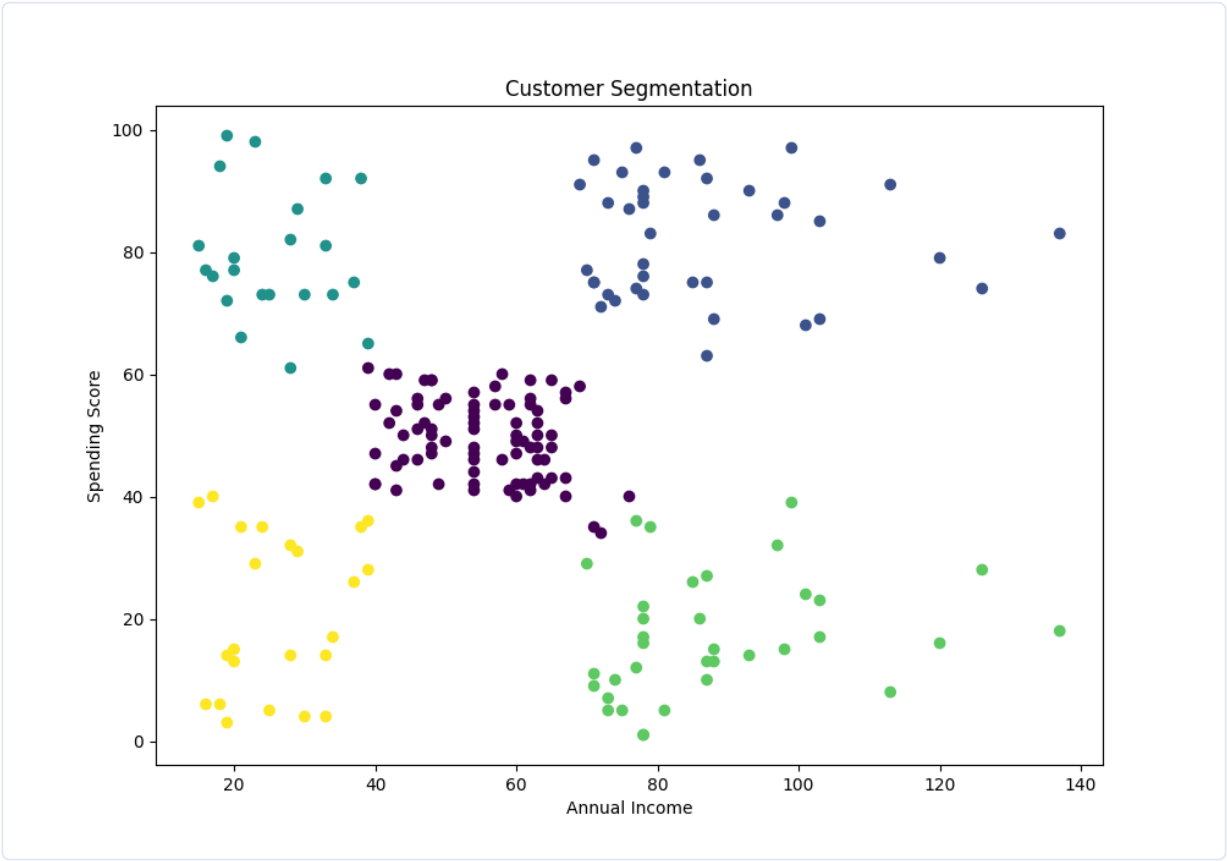


Figure 1: Customer distribution clusters derived via K-Means modeling

6. The 5 Unique Customer Personas Defined

1. **Low Income, Low Spending Score (Cluster 1):** High-frugality consumers who strictly prioritize budget limitations and essential utilities.
2. **Low Income, High Spending Score (Cluster 2):** Highly active trend-driven spenders who display high mall loyalty despite lower baseline annual income.
3. **Average Income, Average Spending Score (Cluster 3):** Standard middle-tier generalist profiles. These compose the largest, most consistent anchor base of the mall.
4. **High Income, Low Spending Score (Cluster 4):** High-income individuals who remain conservative or highly discerning in their purchases. They require exclusive targeted outreach.
5. **High Income, High Spending Score (Cluster 5):** Premium tier target audience. High purchasing power paired with maximum spending metrics makes this segment optimal for luxury product rolling.

7. Core Business Recommendations

- **VIP Retention:** Launch premium privilege tier cards and early-access catalogs to the High Income, High Spending segment.
- **Engagement Triggers:** Deploy limited-time discount strategies to incentivize the High Income, Low Spending group.
- **Feature Expansion:** Future updates to this machine learning pipeline will incorporate Age and Gender categorical metrics to unlock 3D cluster matrices for multi-tier demographic strategies.